# Lightweight Transfer Learning for Segmentation Head Adaptation in Multi-Class Brain Tumor Classification Models

1st Christian Most Tazon
*Escuela Técnica Superior de Ingeniería Informática*
*Universidad Politécnica de Madrid*
Madrid, Spain
christian.most@alumnos.upm.es

2nd Pablo Macías Bradshaw
*Escuela Técnica Superior de Ingeniería de Sistemas Informáticos*
*Universidad Politécnica de Madrid*
Madrid, Spain
pablo.macias@alumnos.upm.es

3rd Sofía Canogar Liao
*Escuela Técnica Superior de Ingeniería de Sistemas Informáticos*
*Universidad Politécnica de Madrid*
Madrid, Spain
sofia.canogar@alumnos.upm.es

***Abstract*—Most diagnosed brain tumors can be classified into gliomas, pituitary tumors, or meningiomas with gliomas accounting for 75% of malignant brain tumors in adults. It is often challenging to make an accurate diagnosis due to the brain's complex structure, limited accessibility to precise medical imaging tools, and the similarities between different kinds of tumors. Artificial Intelligence (AI) and deep learning models can aid diagnosis, particularly in low-resource settings where medical professionals can be lacking, by achieving clinical-level accuracy comparable to neuroradiologists. However, training of new AI tools can be expensive and not feasible in these low resource environments. In this work, we develop a lightweight deep learning model using a pre-trained version of ResNet-18 to classify MRI scans into glioma, meningioma, pituitary tumors, or no tumor. We extend the model with a task-specific segmentation head, training only the new head from scratch while preserving the pretrained classification capabilities. The classification model reached 99.47% classification accuracy. Despite being trained on rough tumor masks, the modified model achieved a Dice score of 0.78 while preserving its classification accuracy. This performance was obtained after only 25 training epochs, using patch-based training followed by whole-image fine-tuning, with the updates restricted to the newly added segmentation head. Error analysis shows higher segmentation errors in meningioma and glioma than in pituitary tumors. Our findings show that existing AI tools can be effectively extended with new functionalities by attaching and training new lightweight task-specific heads without needing to retrain the whole model or degrading its existing capabilities. The resulting fast convergence makes this approach suitable for low resource environments where training new large models from scratch is not feasible.**

***Index Terms*—Brain Tumor, MRI classification, Deep Learning, Segmentation, Low-resource learning, Artificial Intelligence**

## I. Introduction

Brain tumors are the most common type of solid tumors diagnosed in adolescents and children, with more than 88000 adults and 5500 children diagnosed yearly in the United States alone. These tumors are associated with a high mortality rate; the relative survival rate 5 years after a malignant is only 35.6% in adults. Notably, gliomas, a kind of aggressive and often malignant brain tumors, accounts for 75% of malignant tumor diagnosis and causes more than 10000 deaths every year in the US[1][2]. This highlights the current shortcomings in treating brain tumors and the importance of an accurate and early diagnosis to guide treatment and improve patient outcomes. Around 80% of brain tumors can be classified into three major types: meningiomas, pituitary tumors, and the aforementioned gliomas[3]. Magnetic Resonance imaging (MRI) is the preferred tool to diagnose brain tumors[4], with standard imaging protocols demonstrating a high diagnostic accuracy [5]. However, despite the advantages of MRI, definitive diagnosis remains challenging due to the brain's complex anatomy, limited access to high quality imaging and overlapping characteristics of different tumor types[2][6].

Brain tumor diagnosis and classification using deep convolutional neural networks has emerged as a powerful tool to do automatic diagnosis or support neuroradiologists, with segmentation models proven to help neuroradiologists improve their diagnostic accuracy by 12.0 percentage points[7]. Automated brain tumor segmentation models can learn complex spatial features directly from MRI data, largely replacing older handcrafted methods and achieving state of the art results, effectively handling intricate tumor structures that were challenging using traditional approaches[8]. Transfer learning, where models are pre-trained on large-scales datasets such as ImageNet before being adapted to medical imaging tasks, has proven highly effective[9]. Building upon the original U-Net architecture, advances such as Attention U-Net incorporate attention gates to refine the delineation of tumor boundaries, further enhancing performance, reaching a Dice score of 0.807 on the BraTS 2020 dataset, though at the cost of increased computational demands[10]. Despite advances, segmentation remains a resource intensive task, as training can require substantial computational power, specialized hardware, and optimization strategies to achieve practical runtimes[11].

In contrast, deep learning classification models have been widely adopted in medical imaging[12], with lightweight classification models being feasible even in low resource environments[6]. Lightweight architectures, such as MobileNet or EfficientNet, provide high accuracy while requiring limited computational resources[13][14], highlighting a practical gap between feasible classification tools and resource-heavy segmentation models.

Fine-tuning new models by freezing early layers and only training the remaining layers on target allows a model to adapt learned features to a new task without learning from scratch. Such approaches reduce the computational complexity of deep neural network training, proving to be particularly advantageous for medical imaging tasks in low resource environments[15]. In the context of segmentation, models that use a pre-trained encoder backbone to train a decoder have been shown to achieve superior performance compared to models trained entirely from scratch[16].

To address the computational challenge associated with training deep convolutional networks, this paper explores an alternative approach: extending existing trained models with new task-specific heads rather than training entirely new models from scratch. Specifically, we investigate the development of segmentation models, which typically is a resource-intensive task, by expanding more widely available lightweight classification models. By using transfer learning and taking advantage of the learned semantic information in already available classification models, this approach aims to reduce the computational burden of training new segmentation tools, enabling their deployement in low-resource settings where training the state-of-the-art architectures from scratch may not be a feasible solution. Hola

The main contributions of this paper are:

- **A task-specific head transfer learning strategy.** We propose a computationally efficient framework that extends lightweight classification models with a task-specific segmentation head, enabling segmentation capabilities without retraining the full network.
- **Preservation of existing model capabilities.** The proposed approach adds a segmentation functionality while keeping the pretrained classification backbone frozen, retaining its previously learned classification performance.
- **Efficient training for resource-constrained settings.** By restricting training to the newly added segmentation head, the proposed method significantly reduces training costs and increases convergence speed compared to training models from scratch, making it suitable for low-resource environments.
- **Practical validation on brain tumor MRI segmentation.** We domonstrate the effectiveness of the proposed framework on real brain tumor MRI data, showing that a competitive segmentation performance can be achieved even when using rough tumor annotations and limited training epochs by using the proposed framework.

## II. Datasets

### A. Classification Datasets

For the developement of the classification model used for the baseline of this study, a public Dataset of 7023 brain MRI images was used[17]. This dataset was used by Gorenshtein et. al in [6] fot the generation on classification models in a way that is going to be used as the baseline of this project. The Dataset was created by merging three smaller datasets. In total, this dataset contains 1621 glioma tumor images, 1645 meningioma tumor images, 1757 images for pituitary tumors and 2000 non-tumor MRI images. The goal for the baseline is to develop a classification model capable of capturing tumor-indicative patterns for accurate automated detection and diagnosis, while learning representations that can be later used for segmentation. The dataset includes scans spanning a wide range of types, presentations and origins, supporting the models generalizability and aplicability across diverse clinical settings.

### B. Segmentation Datasets

This study developed a brain tumor segmentation model using a Dataset comprised of 4384 MRI images and masks sourced from public datasets, which are divided into tumor classes (glioma, meningioma and pituitary tumor)[18]. There are 1108 glioma scans, 1416 meningioma scans and 1860 pituitary tumor scans with a size of 512x512 pixels. The images require preprocessing due to a non-zero background intensity, with background pixels having a value of 3 instead of 0. The objective of this dataset is to train a segmentation model's decoder to recover spatial information from the semantic information given by a trained encoder.

## III. Baseline Lightweight Classification Model

To investigate whether segmentation models can be efficiently derived from existing lightweight classifiers, this works first establishes a classification baseline designed for low-resource environments. Specifficaly, we adopt the training strategy proposed by Gorenshtein et al.[6], who demonstrated that accurate brain tumor classification models can be obtained via the use of transfer learning with limited computational resources. This approach makes the developement of accurate lightweight classification models feasible for low-resource clinical settings. In this study we replicate and build upon this metodology to construct the classification model that serves as the baseline for our subsequent investigation into extending existing classification models with task-specific segmentation heads.

To replicate this baseline we first initialized a ResNet18 model pre-trained on ImageNet and modified its final FC layer to output four classes, corresponding to glioma, meningioma, pituitary tumor, and no tumor, instead of the original 1000 ImageNet classes. Then the model was trained using the dataset referenced in II-A. First we only trained the newly instantiated FC layer and froze the rest for five epochs and finally we unfroze the last convolutional layer with a lower learning rate and trained the model for another five epochs. To do this we used the standard ImageNet normalization (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]) and size (224x244), as described in [6]. This is done in order to make the images more similar to the data the model has been pre-trained with.

Using this approach we achieved a testing accuracy of 99.24%, closely matching the 98.86% reported by Gorenshtein et al.[6].

## IV. Preparing the Classification Backbone for Segmentation

Due to the preprocessing applied during the classification model training, which applied ImageNet like normalization and resizing, the resulting model is not directly suitable to for MRI segmentation tasks, which typically require domain-specific transformations. Unlike with classification models, MRI segmentation models commonly employ domain-specific preprocesing steps, such as intensity normalization and spatial standarization, to handle MRI-specific characteristics and mitigate scanner variability[19]. To allow for effective patch-based data augmentation, images were resized to 512x512 instead of the 224x224 used in ImageNet preprocessing, preserving suficient spatial resolution for patch extraction given that the ResNet18 model reduces feature map dimensions by a factor of 32.

When applying these data preprocessing steps (size of 512x512 and intensity normalization) instead of the transformations applied during training, the classification model's accuracy colapses, reaching and overall testing accuracy of 30% by just predicting everything as no tumor.

Therefore a step of finetuning is needed to adapt the current model to the preprocessing steps that are going to be used during segmentation. We did this adaptation in two steps: First we only changed the image size to 512x512 but still used ImageNet-like mean and std normalization and adapted the model to this for 5 epochs. Then we also applied an intensity normalization and trained it for 5 more epochs.

After this finetuning process the model reached a testing accuracy of 99.47% with the updated preprocessing, achieving 100% accuracy for tumor vs no tumor classification, as shown in Fig. 1.

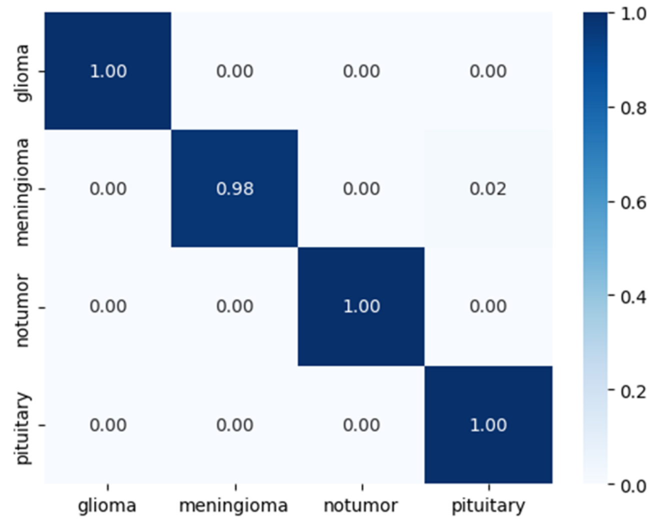


Fig. 1. Normalized confusion matrix.

| Class | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| Glioma | 0.9967 | 0.9967 | 0.9967 | 300 |
| Meningioma | 0.9967 | 0.9804 | 0.9885 | 306 |
| Pituitary | 1.00 | 1.00 | 1.00 | 405 |
| No Tumor | 0.9836 | 1.00 | 0.9917 | 300 |
| Total: | | | | 1311 |
| macro avg | 0.9942 | 0.9943 | 0.9942 | |
| wighted avg | 0.9947 | 0.9947 | 0.9947 | |

Fig. 2. Per-class performance metrics.

## V. Model Architecture: Head-Based Segmentation Extension

To efficiently leverage the pre-trained ResNet18 backbone we propose a head-based extension that adds a lightweight decoder for segmentation while preserving the encoder's learned features. This design enables effective reuse of the encoder's learned representations while restricting training to the decoder, thereby significantly reducing the computational cost of training.

We based the design of our custom architecture on the U-Net architecture by adding a U-Net-like decoder[20][16] with skip-connections to the encoder; see Fig. 3. This architecture is designed to take full advantage of the encoder's learned features by introducing skip-connections at every semantical level, combining encoder feature maps with the upsampled outputs of the decoder at corresponding semantic levels. With this we try to make it as easy as possible for the decoder to converge without having to modify the encoder. The ResNet18 backbone of the architecture remains completly unchanged, mantaining all of its learned features as well as its original classification capabilities. As, in contrast with the original U-Net architecture, ResNet18 halves the spatial size of the feature maps with its first layer, upsampling is needed at the end of the decoder in order to bring the output mask to the same spatial resolution as the input image. For this we used bilinear interpolation.

Overall, the architecture has two task-specific output heads: a classification head, which outputs a class probability distribution, and a segmentation head, which outputs the predicted tumor mask.

In order to reduce unnecessary operations when training for only a specific task we added "modes" to our model, changing

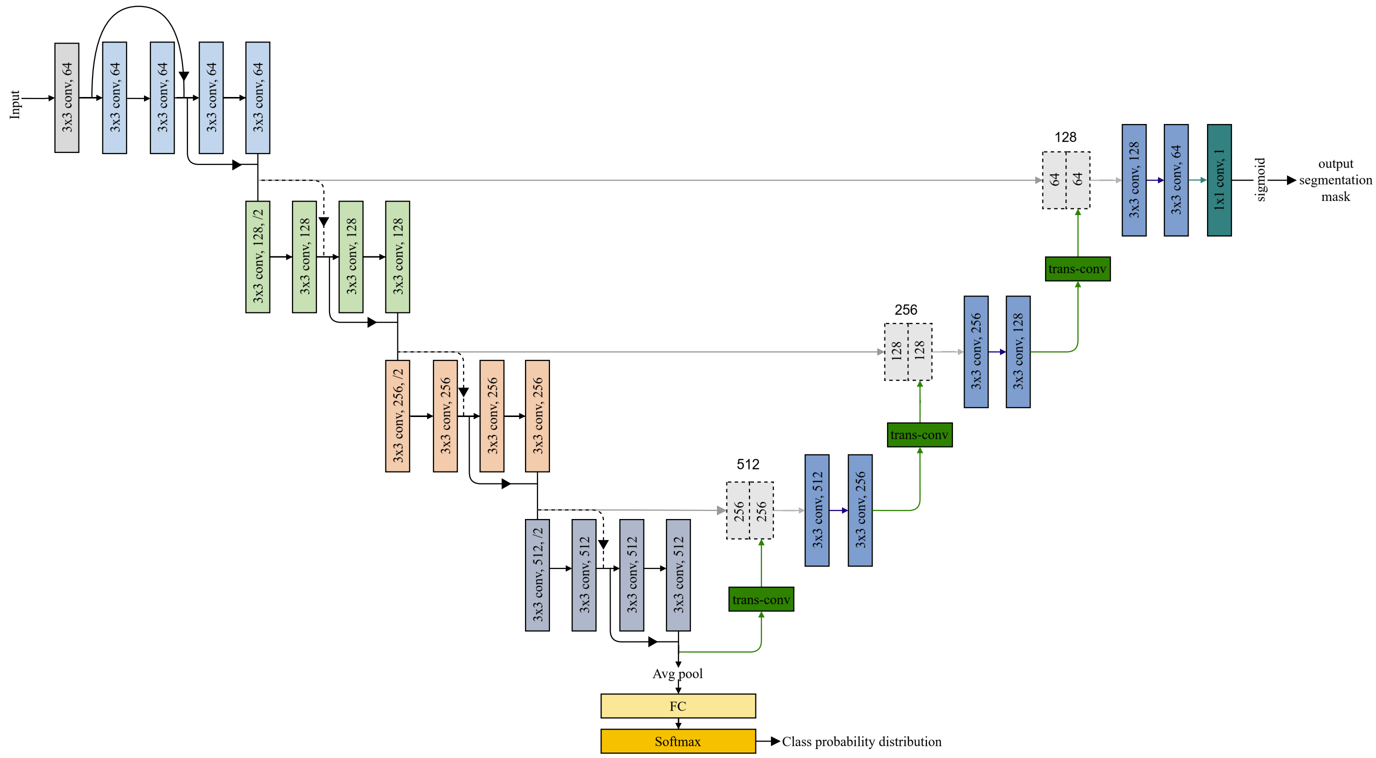


Fig. 3. Overview of the proposed head-based segmentation architecture with a ResNet18 encoder and U-Net-like decoder.

the total output by turning off one of the heads, making the model have one output instead of two when needed.

## VI. Training Strategy

To train the newly instantiated decoder we used the dataset referenced in II-B. All MRI-scans used underwent a preprocessing and augmentation pipeline where we applied intensity normalization and resized all images to a spatial resoluton of 512x512. Prior to normalization background pixel values were corrected, as the dataset's background pixels had an intensity value of 3 (on a 0 - 255 scale) instead of 0.

The training is divided into two parts: patch training and whole-image training. During the first phase we also applied a data agumentation step to the preprocessing pipeline: We generated 6 randomly cropped patches with a size of 288x288, from which at least 70% contained a part of the tumor. This strategy encourages the model to focus on tumor-specific features and how the tumor affects the directly sorounding brain tissue, while reducing time spent learning trivial background representations. For validation during this phase sliding-window interface with 50% patch overlap is used.

During the second phase we finetune the model using whole images in order to help the model learn global characteristics about the tumor, like its location and how it shapes all of the tumor tissue arround it, and to reduce false positives caused by patch-level context limitations.

The training was done over a total of 25 epochs, 10 epochs using patch-training and 15 epochs with whole-image-training, during which the encoder was kept frozen.

## VII. Results

## VIII. Discussion

## IX. Conclusion

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